

Data Cleaning & PCA

Exploratory Analysis of Hospital Patient Dataset

Research Question

Is there a correlation between patient income levels and number of days initially spent in the hospital? i.e. are higher income patients staying longer for treatment?

Dataset Variables

CaseOrder: The order in which patients were admitted for treatment. This is a qualitative chronological order value. For example 1, 50, 78.

Customer_ID: This column is a unique identifier for all patients or visits and is a qualitative string as values such as C412403 include numbers and letters.

Interaction: This column is used to differentiate any patient visit from their other visits, a qualitative string of numbers and letters, i.e. 8cd49b13-f45a-4b47-a2bd-173ffa932c2f

UID: This column is a unique identifier for patients or visits, and is a qualitative string of numbers and letters, i.e. 3a83ddb66e2ae73798bdf1d705dc0932

City: This column shows city location in qualitative string form, as there are too many cities to categorize. i.e. Eva, Marianna

State: This column state location in categorical form, as there are only 50 options and must fit the two letter format. i.e. AL, TX

County: This column shows county location in qualitative string form, as there are too many counties to categorize. i.e. Morgan, Jackson

Zip: This column is a 5 digit location identifier in qualitative form, as values are too large of a dataset to categorize. i.e. 35621, 32446

Lat: Location identifier as numeric values in decimal. i.e. 34.3496, 30.84513

Lng: Location identifier as numeric values in decimal. i.e. -86.7251, -85.2291

Population: Quantitative indication of area's human population, in integer form as these must be whole numbers. i.e. 2591, 11303

Area: Categorical indicator of type of environment one resides in. ie. Suburban, urban

Timezone: Categorical indicator of timezone location. i.e. America/Chicago, America/new_york

Job: Patient's occupation in qualitative string form as this can take many differing inputs. i.e. psychologist, sport and exercise, chief executive officer

Children: Quantitative value of patient's child count. i.e. 1, 2.

Age: Quantitative value of patient's age in years. i.e. 53, 51

Education: Categorical indication of how much school patient has completed. i.e. some college, regular high school diploma

Employment: Categorical indication of employment status. i.e. full time, retired

Income: Quantitative value of patient's annual income. i.e. 86575.93, 46805.99

Marital: Categorical indicator of marriage status. i.e. divorced, married

Gender: Categorical indicator of gender. i.e. Male, Female

ReAdmis: Categorical value indicating if patient was readmitted to hospital after initial treatment. i.e. yes, no

VitD_levels: Quantitative value of measured vitamin D level in patient. i.e. 17.80233, 18.99464

Doc_visits: Quantitative value of patient's number of doctor visits. i.e. 6, 4

Full_meals: Quantitative value of patient's number of full meals eaten per day, self reported. i.e. 1, 4

vitD_supp: Categorical indication of how much vitamin D support each patient currently uses. i.e. 0, 1

soft_drink: Categorical indication as to whether patient consumes soft drinks regularly or not. i.e. Yes, No

initial_admin: Categorical indication as to how patient arrived at hospital. i.e. emergency admission, elective admission

HighBlood: Categorical indication for patient's high blood pressure. i.e. yes, no

Stroke: Categorical indication as to if patient has suffered a stroke in their history. i.e. yes, no

Complication_risk: Categorical indication of risk of danger in medical treatments for patient due to prior factors. i.e. high, medium

Overweight: Categorical indication of patient's weight class. i.e. 0, 1

Arthritis: Categorical indication of patient's Arthritis status. i.e. yes, no

Diabetes: Categorical indication of patient's Diabetes status. i.e. yes, no

Hyperlipidemia: Categorical indication of patient's Hyperlipidemia status. i.e. yes, no

Backpain: Categorical indication of patient's presence of back pain. i.e. yes, no

Anxiety: Categorical indication of patient's presence of anxiety. i.e. 1, 0

Allergic_rhinitis: Categorical indication of patient's presence of allergic rhinitis. i.e. yes, no

Reflux_esophagitis: Categorical indication of patient's presence of reflux esophagitis. i.e. yes, no

Asthma: Categorical indication of patient's presence of Asthma. i.e. yes, no

Services: Categorical indication of which services patient was administered. i.e. blood work, CT scan

Initial_days: Quantitative value of how many days patients spent in hospital for initial visit. i.e. 10.585

Totalcharge: Quantitative value of each patient's total bill upon leaving hospital. i.e. 3191.048774

Additional_charges: Quantitative value of additional billing amounts to be sent to patients. i.e. 17939.40342

Item1: Patient's opinion on their hospital experience's Timely admission on a scale of 1-8

Item2: Patient's opinion on their hospital experience's Timely treatment on a scale of 1-8

Item3: Patient's opinion on their hospital experience's Timely visits on a scale of 1-8

Item4: Patient's opinion on their hospital experience's Reliability on a scale of 1-8

Item5: Patient's opinion on their hospital's Options on a scale of 1-8

Item6: Patient's opinion on their hospital's Hours of treatment on a scale of 1-8

Item7: Patient's opinion on their hospital's Courteous staff on a scale of 1-8

Item8: Patient's opinion on their hospital experience's Evidence of active listening from doctor on a scale of 1-8

Data Cleaning Plan

This data set should be assessed for duplicate values, missing values, outliers, and re-expression of categorical variables. Duplicate values will be located using `.duplicated()` and should be removed from the data set. Missing values will be evaluated using `msno.matrix()`, but no rows will be dropped due to missing data, we will just impute appropriate values as to not allow missingness to interfere with our calculations. Outliers will be assessed using seaborn's `.boxplot()`, and then will be replaced with the median value using `.median()`. All categorical variables will have their category set reviewed to ensure that any re-expressed categories are condensed appropriately. For binary columns of yes/no or 1/0, we will convert all 1/0 fields into yes/no for simplicity. Then any NA values will be replaced with 'No', as it would be prudent to not assume any patient has a disease without confirmation.

Data Quality Assessment

Nullity has been assessed using the `.isnull()` function for all variable types, which showed us all rows that need to have null values dealt with, and we have now done so for each row. Python's `.duplicated()` function is perfect for analyzing duplicates of all data types, as it can instantly parse all rows and return any where all of the data is identical to any other. This returning no results confirms that each row is indeed a unique data point. Using seaborn's boxplot function is a perfect way to analyze outliers on any numeric variables, as it allows for the numeric count of outliers, along with showing the entire range within which values fall both inside and out of the boxplot. Using the `.unique()` function on all categorical variables showed us the set from which each category can derive, giving us confidence that no other answers can possibly exist. Eliminating inconsistency from binary columns will allow for easy understanding of our data at a glance, and any NA values should default to just not having that condition.

I will go about cleaning this data set using the pandas package inside of Python, as the pandas package will easily allow this csv file to be converted into a data frame that can easily sum up null, missing, and outlying values. This should smoothly allow for an understanding of what the data is missing or has as an invalid value. Pandas will also allow for simple functions like `.replace`, `.unique`, `.fillna` that we will use. The missingno package will also be used in order to visualize the missingness of this dataset. I will also import seaborn in order to use the boxplot function to analyze outliers. Numpy will also be imported in order to do simple comparisons inside of the `.where()` function, and to then impute those specified values with nan to

allow for easy imputation of chosen values. The packages pandas, seaborn and missingno are the key benefits of Python over R for this exercise. I will use sklearn.decomposition and matplotlib in order to perform PCA after the data has been cleaned.

Findings: Missing Values

`df_nullity.mean() * 100` shows us how much of each field we are missing. We can see that many columns have a significant amount of missing data. Most of the missing fields look like things patients didn't find crucial to provide for treatment such as age, number of children, and income. I believe this data to be missing completely at random. However, we can see that all of the fields provided by the hospital side such as services, TotalCharge, and Additional_charges are present. While we don't have meaningful patterns in the missing data useful to impute these values, we can conclude that the hospital side is much more accurate at getting each field filled out while patients are more likely to skip fields they don't view as crucial for treatment.

Findings: Duplicates

To assess duplicate values, we will use the `.duplicated()` function. This returns False with a count of 10,000 (which we know is how many rows are in the data set), telling us that we have no duplicated values to handle in this dataset.

Findings: Numeric Column Outliers/NAs

To assess outliers, we will create a boxplot for each numeric variable and visually assess where outliers begin, in addition to counting the amount of outliers by subtracting the amount of NON outliers from the total number of rows in the data set. Any outlying values will be replaced with NA, and then all NA values in each column will be imputed with an appropriate replacement value.

We can subtract the 7206 number we see here out of the 10000 total rows in our data set, giving us 2794 outlier values. We can visually see that the outliers begin at the value of \$100,000, so we cut off there and replace outliers above that value with the median. The income boxplot ranges from \$0 - \$100,000.

For total charge, we can see that outliers begin at the value \$14,000, so we replace the 472 outlying data points above that value. The total charge boxplot ranges from \$0 to \$14,000.

For additional charge, we can see outliers begin above \$27,000, with 432 such values. This data needs to be converted to float to calculate with consistency. The additional charges boxplot ranges from \$0 to \$27,000.

The initial days boxplot shows no outliers, so there will be no adjusting outliers. However, we saw earlier that this column has some null data, and those values will be imputed with the median.

Vitamin D levels has outliers on both ends of its boxplot, below the value 11, and above the value 24, as the boxplot ranges from 11 to 24. We can see there are 528 outliers above the value 24, and 10 outliers below the value 11. All outlying values will be imputed with the median.

The boxplot for age shows no outliers. We replace NA with the median and do not adjust outliers. The age boxplot ranges from 18 to 89.

The boxplot for number of children shows a range from 0-7, with 3 values outlying above 7. These do not need imputed, but we will impute nulls with the median.

The population boxplot shows that outliers begin at the value 26,000, with a boxplot ranging from 0 to 26,000. However, I do not believe it prudent to impute any of these values, as high population areas are an important factor in medical studies. So we replace any NA values with the median and do not adjust outliers.

The doctors visits boxplot ranges from 1-9 and has no outliers outside of this range. We will impute NA values with the median and not adjust outliers.

The full meals eaten boxplot ranges from 0-5 with two outliers above the value 5, these can be left as is, but we will impute any null values with the median.

Findings: Categorical Re-expression

For categorical re-expression, we will analyze the unique values present in categorical columns and look for redundancies/typos. All of these category sets seem to be complete and have no redundancies or typos, so all of these columns are fine and do not need categorical re-expression adjustments.

For the binary columns, we will use `.replace` to change all 1/0s to yes/nos. This is done for anxiety and overweight. Then any NA values will be taken care of using `.fillna`. I found that anxiety and overweight had 'nan' as a string, and soft drink had true NA values. I used the code that would take care of both instances of Na for all three columns in need of cleaning.

Justification of Methods

Firstly, we have imputed null values using the median, mean, or appropriate default value from the data set. I now used the `.duplicated()` function to prove that no duplicates exist in the data set. Outliers have been assessed for each numeric variable, and then replaced with the median value for that variable, this allows us to work with more common relevant data and removes the least relevant, least useful data. The `.unique()` function confirms that each categorical variable does not contain re-expression, but shows us all possible categories, giving us confidence that all fields are properly accounted for. I've used the `.replace` function to replace 1/0 with yes/no, and this change was confirmed using `.unique` on those columns, showing us there are only yes/no as possible answers now. These steps result in a data set that has converted any null values into a reasonable estimate, removed all outlying values, ensured all categories of variables are accurate, and has no duplication. Now this data can be analyzed as we have created consistency and removed issues that would prevent calculations or skew the results.

Cleaning Outcomes

Looking for duplicates did not result in a change in the data, but we know it is more meaningful and accurate because of that. Removing outliers takes out the least common occurrences of data, essentially removing

the least likely instances from our calculations. The data set is conveniently condensed, easier to visualize, and we will not have skewed general results from a few highly unlikely values. We have also taken any null values that would halt calculations and have imputed them with the median so we can run calculations on all columns. Any numeric columns were also converted to float form in order to allow for calculations and consistency. Re-expression of categories also did not result in a tangible change, but knowing that these are accurate allows us to draw conclusions using those fields moving forward. Binary columns are now consistent with all yes/no, and all previously NA fields have filled with a No answer.

Limitations

The largest limitation of this process is the fact that it is not possible to confirm whether these data points are accurate, does the patient have a biased answer? Did the patient not answer? How trained and how much attention can the doctors spare on getting this data in, are they rushed? Many fields are missing, and typos happen all the time. This is part of why we remove outliers, but all incorrect fields can never be fully corrected. Doctors and patients make mistakes, typos, and can be incorrect. These just have to be mitigated by having a large enough data set.

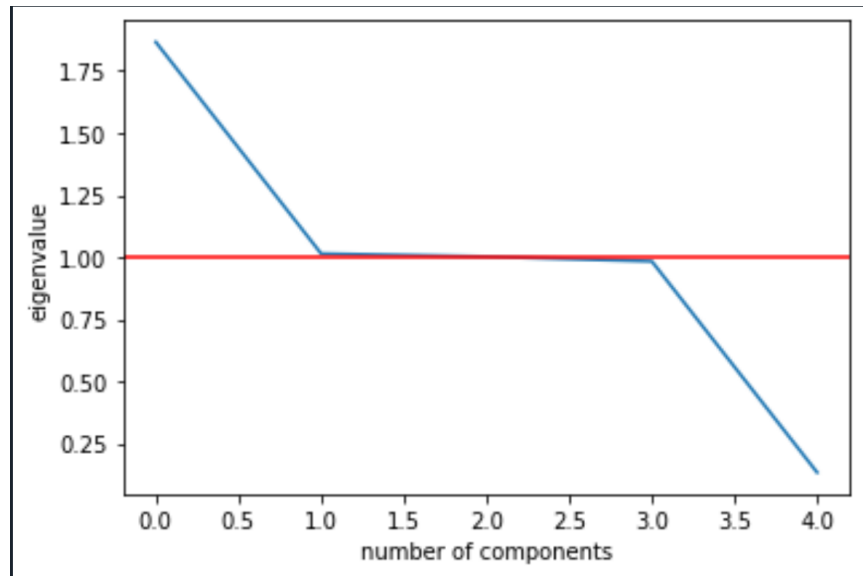
The largest result from the limitations of our missing or inaccurate data is that the columns that are missing data will not be able to be analyzed for PCA nearly as well as the fully completed columns. Only a few columns have missing data, but each are missing a significant chunk. Any conclusions drawn from columns that have high nullity such as overweight, age, number of children, will be much weaker conclusions that have less of a data set evidencing the conclusion. There could also be potential skews as to the pattern of missing data, such as overweight patients being more likely to not answer the overweight column, people with no children not answering how many they have, or similar situations leading to data that just goes unanswered by a significantly skewed proportion of our population. We do not understand exactly how these would work.

PCA: Principal Components

There are 5 numeric variables taken from the medical data information, so there will be 5 principal components.

A	B	C	D	E	F	G
	PC1	PC2	PC3	PC4	PC5	
Income	-0.00831	0.022025	0.933711	-0.35725	-0.00164	
VitD_levels	0.045283	0.707441	0.237158	0.662617	-0.0466	
Initial_days	0.70526	-0.06891	-0.00229	-0.02341	-0.7052	
TotalCharg	0.70714	0.002766	0.006049	-0.00372	0.707033	
Additional_	0.021034	0.703055	-0.26814	-0.65784	-0.02495	

PCA: Scree Plot & Reduced Components



This plot shows that there are three components because the eigenvalues dip below the red line representing eigenvalue of 1, at principal component index two, as python defaults to a 0 index, this represents three total principal components with an eigenvalue greater than one, which tells us to retain three principal components.

PCA: Organizational Benefit

The results derived from the above PCA would be able to benefit the company by allowing for a simple visualization of many combined values from the original data set, this allows us to take conclusions from our graph such as how many PCA components are necessary and how impactful they are. From that point it is possible to select the features that have the most impact on any future calculations. This will also 'reduce noise' as the least relevant factors will no longer contribute data points. This can also have the potential benefit of speeding up machine learning algorithms, but that is likely not the priority of PCA here. In the example of PCA here, this would allow the organization to lower their PCA components from five to three. This should simplify any calculations, decrease processing time, and allow for an understanding of which factors are most relevant and impactful for the company.